



1 Mark for Review



Note: Figure not drawn to scale.

The triangle shown has one side with a length of x inches and two sides each with a length of y inches. The perimeter of the triangle is 47 inches. Which equation represents this situation?

(A) $x + y = 47$

(B) $x + 2y = 47$

(C) $2x + y = 47$

(D) $2x + 2y = 47$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	$-1/3$ -.3333 -0.333	-.33 -0.33

2

Mark for Review

The function m is defined by $m(x) = 5x + 3$, and the function p is defined by $p(x) = 3 - x$. What is the value of $2m(3) - p(3)$?

Answer Preview:

Telegram @satuzbhunters

3 Mark for Review



A batch of smoothies consists of 4 cups of milk and 2 bananas and has 1,228 milligrams (mg) of calcium. There is 304 mg of calcium in 1 cup of milk. How much calcium, in mg, is in 1 banana?

(A) 1,216

(B) 462

(C) 12

(D) 6

Telegram @satuzbhunters

4

Mark for Review



A furniture company is selling all tables with a 20% discount off their original price for the month of October. What is the discount, in dollars, on a table with an original price of \$130?

(A) 110

(B) 90

(C) 52

(D) 26

5.

Find the factors of

$$48x^4 + 200x^2 + 208$$

in the form $ax^2 + b$, where a and b are integers.

Then, find the minimum value of ab .

6

Mark for Review



$$4x = 16$$

$$-3x + y = -5$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

(A) -21

(B) -11

(C) 11

(D) 21



Hide

Calculator

Reference

More

52%

7 Mark for Review



A shopper is purchasing oranges that cost \$0.67 each. If the shopper has a coupon for \$2 off the entire purchase and there is no sales tax, which inequality represents the number of oranges, x , that the shopper can purchase for no more than \$12?

(A) $0.67x + 2 \leq 12$

(B) $0.67x + 2 \geq 12$

(C) $0.67x - 2 \leq 12$

(D) $0.67x - 2 \geq 12$



Telegram @satuzbhunters



Hide

Calculator

Reference

More

52%

8 Mark for Review



Circle T has a radius of 35 millimeters (mm). What is the area of circle T , in mm^2 ?

(A) 35π

(B) 70π

(C) 140π

(D) $1,225\pi$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable; will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67

9

Mark for Review

In the xy -plane, circle F is defined by the equation $(x + 7)^2 + (y + 7)^2 = 36$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 7)^2 + (y + 7)^2 = p$, where p is a constant. What is the value of p ?

Answer Preview:



Hide

Calculator

Reference

More

51%

10

Mark for Review



Which expression is equivalent to $4yz^3 + 5yz + 3yz^3$?

(A) $7yz^3 + 5yz$

(B) $7yz^6 + 5yz$

(C) $12yz^6$

(D) $12yz^7$



- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

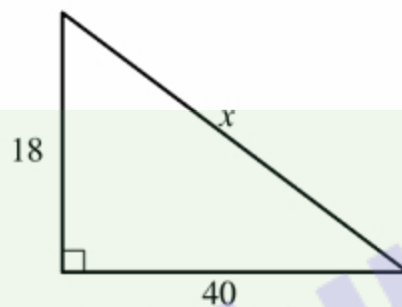
11

Mark for Review

If $\frac{x}{y} = 76$ and $\frac{cx}{4y} = 76$, what is the value of c ?

Answer Preview:

Telegram @satuzbhunters

12 [Mark for Review](#)

Note: Figure not drawn to scale.

In the right triangle shown, what is the value of x ?

(A) $\sqrt{58}$

(B) $\sqrt{116}$

(C) $\sqrt{720}$

(D) $\sqrt{1,924}$



$$3x = 45y - 27$$

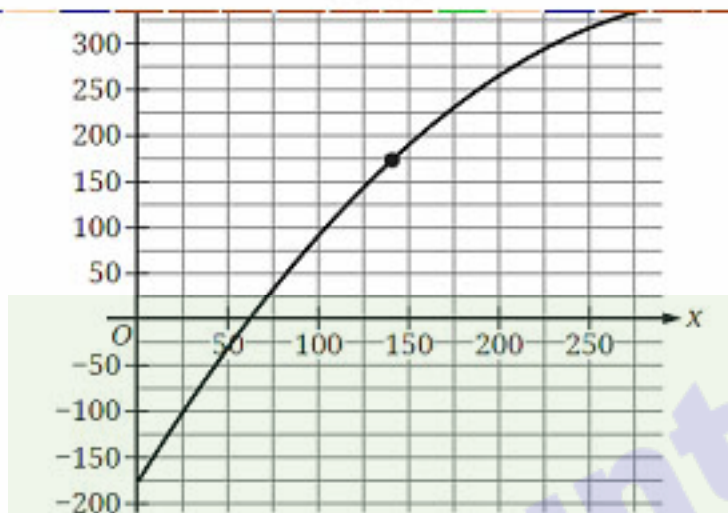
One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

(A) $x = 5y$

(B) $\frac{1}{3}x = 5y$

(C) $x = 15y - 9$

(D) $\frac{1}{3}x = 15y - 9$



The graph shows the estimated boiling point y , in degrees Celsius, of a straight-chain alkane with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(141.32, 172.26)$?

- (A) A straight-chain alkane with a molecular weight of 172.26 grams per mole has an estimated boiling point of 141.32 degrees Celsius.
- (B) A straight-chain alkane with a molecular weight of 141.32 grams per mole has an estimated boiling point of 172.26 degrees Celsius.
- (C) The minimum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 141.32 grams per mole and an estimated boiling point of 172.26 degrees Celsius.
- (D) The maximum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 141.32 grams per mole and an estimated boiling point of 172.26 degrees Celsius.



Hide

Calculator

Reference

More

48%

15

Mark for Review



If $\frac{72}{x} = 6x$, what is the value of $6x^2$?

(A) 432

(B) 72

(C) 12

(D) 6

16

Mark for Review

ABC

$$y \leq 5x - 21$$

Which point (x, y) is a solution to the given inequality in the xy -plane?

(A) $(-5, -21)$

(B) $(0, 21)$

(C) $(5, -21)$

(D) $(6, 21)$

Telegram @satuzbhunters



A square map has a side length of 55 inches, and 1 inch on the map represents an actual distance of 13 miles. A smaller version of the same map is printed as a square with the side length 70% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

(A) 3.90

(B) 7.65

(C) 38.50

(D) 43.33



1:50

5 minutes left in this part of the test.

18

Mark for Review



$$f = \frac{(10x)^2}{7g}$$

The given equation relates the positive numbers f , x , and g . Which equation correctly expresses x in terms of f and g ?

(A) $x = \frac{\sqrt{7fg}}{10}$

(B) $x = 10\sqrt{7fg}$

(C) $x = \sqrt{\frac{7fg}{10}}$

(D) $x = \sqrt{7fg} - 10$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67

19

Mark for Review

In the xy -plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 4$. If line ℓ also passes through the point $(6, d)$, what is the value of d ?

Answer Preview:



20

Mark for Review



A circle in the xy -plane has its center at $(14, 19)$ and has a radius of $7k$. Which equation represents this circle?

(A) $(x - 14)^2 + (y - 19)^2 = 49k$

(B) $(x - 14)^2 + (y - 19)^2 = 49k^2$

(C) $(x - 14)^2 + (y - 19)^2 = 7k$

(D) $(x - 14)^2 + (y - 19)^2 = 7k^2$

Telegram @satuzbhunters

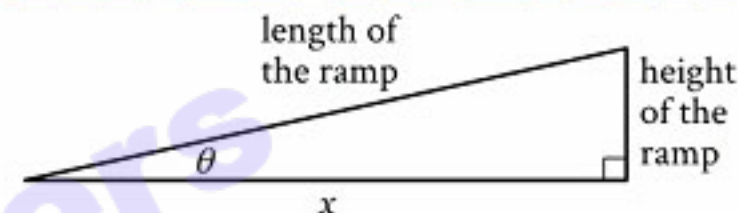
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

21

Mark for Review



Note: Figure not drawn to scale.

According to a US law, ramps for use by the general public must form an angle with level ground such that $\tan \theta \leq \frac{1}{12}$. If the ramp in the figure conforms to this law and has a height of 29.1 inches, what is the least possible value of x , in inches?

Answer Preview:

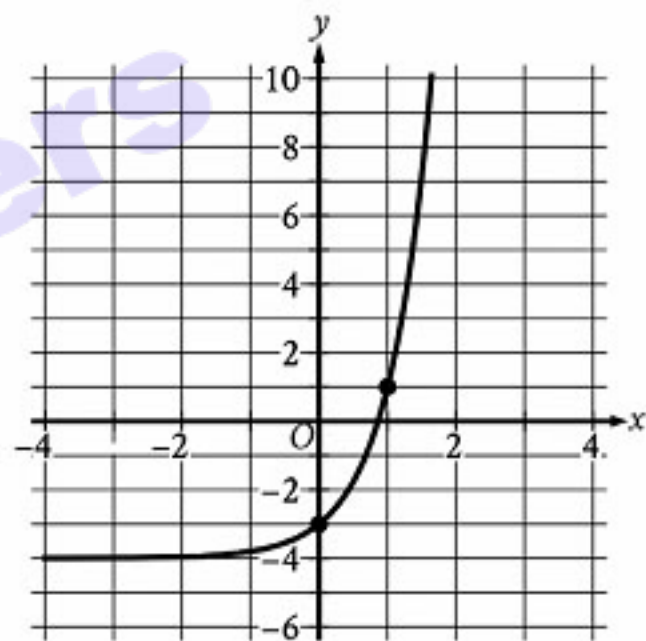
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	$-1/3$ -.3333 -0.333	-.33 -0.33

22

Mark for Review



The equation of the graph shown is $y = a^x + b$, where a and b are constants. What is the value of $a - b$?

Answer Preview:

1 Mark for Review



$$f(x) = \frac{x+9}{3}$$

For the function f defined by the equation shown, what is the value of $f(-6)$?

(A) -3

(B) -1

(C) 1

(D) 3

Telegram @satuzbhunters

2

Mark for Review



28, 28, 29, 30, 32, 33, 37

What is the mean of the data set shown?

☐ (A) 28☐ (B) 30☐ (C) 31☐ (D) 37



Hide



Calculator



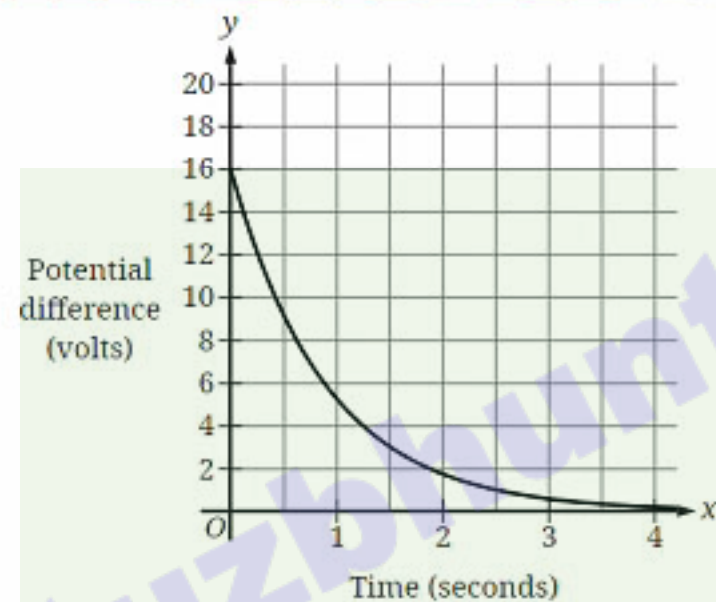
Reference



More

3

Mark for Review



The potential difference across a capacitor in an electric circuit is initially 16 volts and decreases with time, as shown in the graph. Which of the following is closest to the time, in seconds, from the initial measurement at which the potential difference across the capacitor is 15% of its initial value?

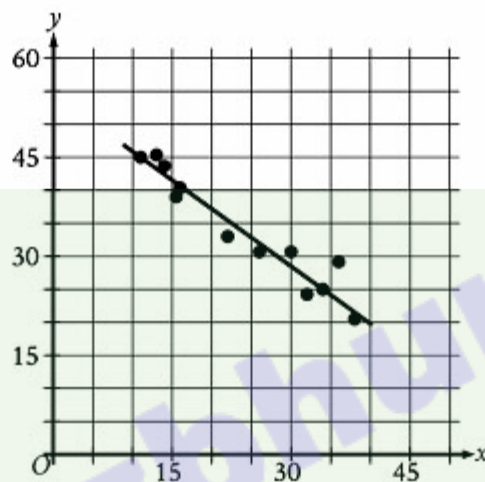
(A) 0.43

(B) 0.85

(C) 1.71

(D) 3.41





The scatterplot shows the relationship between two variables, x and y , for data set A. A line of best fit for the data is also shown. Data set B is created by subtracting 9 units from the value of y for each data point from data set A. Which of the following is closest to the y -coordinate of the y -intercept of the line of best fit for data set B?

(A) 45.57

(B) 53

(C) 59.62

(D) 62.75

5

Mark for Review



In the linear function f , $f(0) = 5$ and $f(7) = 5$. Which equation defines f ?

(A) $f(x) = 0$

(B) $f(x) = 7$

(C) $f(x) = 5$

(D) $f(x) = x + 5$

Telegram @satuzbhunters



Hide

Calculator

Reference

More

41%

6

Mark for Review



x	$f(x)$
1	-20
5	-40
10	-65

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

(A) $f(x) = -5(5x - 3)$

(B) $f(x) = -5(5x - 4)$

(C) $f(x) = -5(x + 3)$

(D) $f(x) = -5(x - 4)$

7  Mark for Review

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 128(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 128-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 32$ in this context?

- (A) Approximately 32 grams of the sample remains 58 years after the sample starts to decay.
- (B) The mass of the sample has decreased by approximately 32 grams 58 years after the sample starts to decay.
- (C) The mass of the sample has decreased by approximately 58 grams 32 years after the sample starts to decay.
- (D) Approximately 58 grams of the sample remains 32 years after the sample starts to decay.

8 Mark for Review



The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	34%
18–40 years old	23%
41–65 years old	22%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

(A) 0.21

(B) 0.32

(C) 0.45

(D) 0.62



- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$\frac{2}{3}$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	$-\frac{1}{3}$ -.3333 -0.333	-.33 -0.33

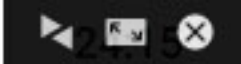
9

Mark for Review

A right circular cone has a base diameter of 28 inches and a height of 6 inches. The volume of this cone is $k\pi$ cubic inches. What is the value of k ?

Answer Preview:

Telegram @satuzbhunters



Hide

Calculator

Reference

More

10

Mark for Review



In triangle ABC , point D is the midpoint of segment AB and point E is the midpoint of segment BC . If $BD = 8$ and $DE = 14$, what is the length of segment AC ?

(A) 28

(B) 26

(C) 22

(D) 16

11

Mark for Review



$$x(r - 7) + 3 = 19x + 25$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

(A) 3

(B) 7

(C) 22

(D) 26

12

Mark for Review



In a collection of items, 10% are red, 25% are green, 35% are blue, and 30% are yellow. If there are 40 green items, how many blue items are in the collection?

(A) 14

(B) 16

(C) 48

(D) 56

Telegram @satuzbhunters



Hide



Calculator



Reference



More

13

Mark for Review



$$x(13x + 27) + c = 5$$

In the given equation, c is a constant. A solution to the equation is $\frac{-27 + \sqrt{261}}{26}$. What is the value of c ?

(A) 14

(B) 9

(C) 5

(D) 4

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67

14

Mark for Review

One of the factors of $2x^3 + 44x^2 + 234x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Answer Preview:



15

Mark for Review



A model estimates that at the end of each year from 2015 to 2020, the number of rabbits in a population was 170% more than the number of rabbits in the population at the end of the previous year. The model estimates that at the end of 2016, there were 216 rabbits in the population. Which of the following equations represents this model, where n is the estimated number of rabbits in the population t years after the end of 2015 and $t \leq 5$?

(A) $n = 80(1.7)^t$

(B) $n = 80(2.7)^t$

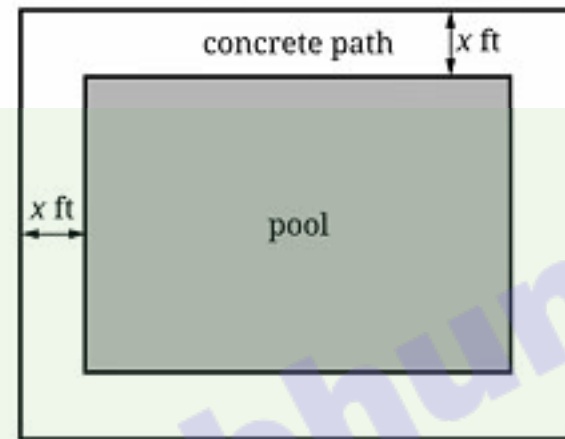
(C) $n = 216(1.7)^t$

(D) $n = 216(2.7)^t$

Telegram @satuzbhunters

16

Mark for Review



Note: Figure not drawn to scale.

The figure shows a rectangular pool surrounded by a concrete path that is x feet (ft) wide on all sides. The pool is 21 ft long and 13 ft wide. The area of the concrete path is 240 ft^2 . What is the value of x ?

(A) 3

(B) 6

(C) 20

(D) 40

17

Mark for Review

ABC

A circle in the xy -plane has its center at $(-1, 1)$. Line ℓ is tangent to this circle at the point $(7, -6)$. Which of the following points also lies on line ℓ ?

(A) $(0, \frac{8}{7})$

(B) $(6, 9)$

(C) $(14, 2)$

(D) $(15, 1)$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67

18

Mark for Review

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

Answer Preview:

19

Mark for Review



To investigate the effect of magnesium supplementation on the sleep quality of adults, a researcher selected 190 adults at random from a community center to participate in a study. The researcher first measured the sleep efficiency of each participant. Each participant was then randomly assigned to take a magnesium supplement or a placebo each day for 6 weeks. At the end of the 6 weeks, the researcher measured the sleep efficiency of all participants again and found that the magnesium supplement caused statistically significant improved sleep quality for the adults at the community center. What feature of this study allowed the researcher to conclude that the magnesium supplement caused the improved sleep quality?

- (A) There were more than 100 participants in the study.
- (B) The participants were selected at random from the community center.
- (C) The sleep efficiency of each participant was measured again at the end of 6 weeks.
- (D) Each participant was randomly assigned to take a magnesium supplement or a placebo.

Telegram @satuzbhunters



Hide

20

Mark for Review



The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

I. $f(x) = a(2.1)^{x+b}$

II. $g(x) = a(2.1)^x + b$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67

21

Mark for Review

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{358})$. If $h(m) = 0$, what is the greatest possible value of m ?

Answer Preview:

Telegram @satuzbhunters

22

Mark for Review



In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 24 and the length of $\overline{AC}$ is 13 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

(A) $\frac{13}{24}$

(B) $\frac{24}{37}$

(C) $\frac{37}{24}$

(D) $\frac{24}{13}$

Telegram @satuzbhunters